



Chapter 5. Signaling systems

Composed by Nguyen Hoang Viet, Bui Thu Cao

Báo hiệu

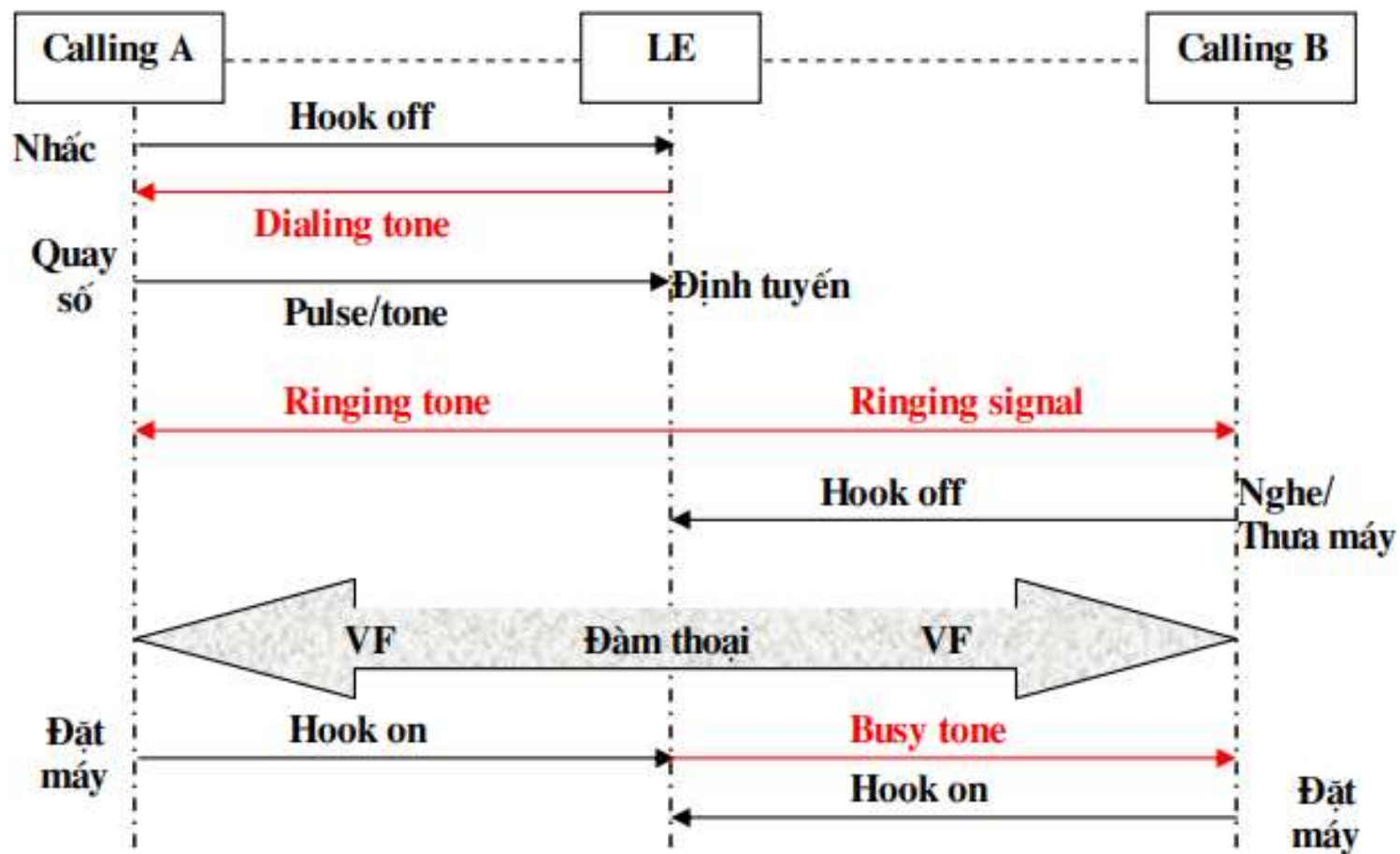
■ Báo hiệu

- Báo hiệu thuê bao
- Báo hiệu kênh liên kết (CAS)
- Báo hiệu kênh chung (CCS)

■ Giới thiệu chung về báo hiệu liên đài

- Hiện nay có nhiều hệ thống báo hiệu giữa các tổng đài, phân chia làm hai loại CAS, CCS (Channel Associated Signaling – Báo hiệu kênh kết hợp và Common Channel Signaling- Báo hiệu kênh chung).
 - CAS: giữa các tổng đài phải có các thông tin báo hiệu chia thành các kênh báo hiệu, mỗi kênh được gán cho một kênh thoại. Thông tin báo hiệu trên kênh báo hiệu đó sẽ cho biết về kết nối liên quan tới kênh thoại được gán kết. Trong một số trường hợp sử dụng các kênh thoại truyền thông tin báo hiệu cho kết nối liên quan tới chính nó.
 - CCS: liên kết báo hiệu giữa hai tổng đài sử dụng chung cho tất cả các kênh thoại nối giữa hai tổng đài đó đó bằng cách ghép gói.

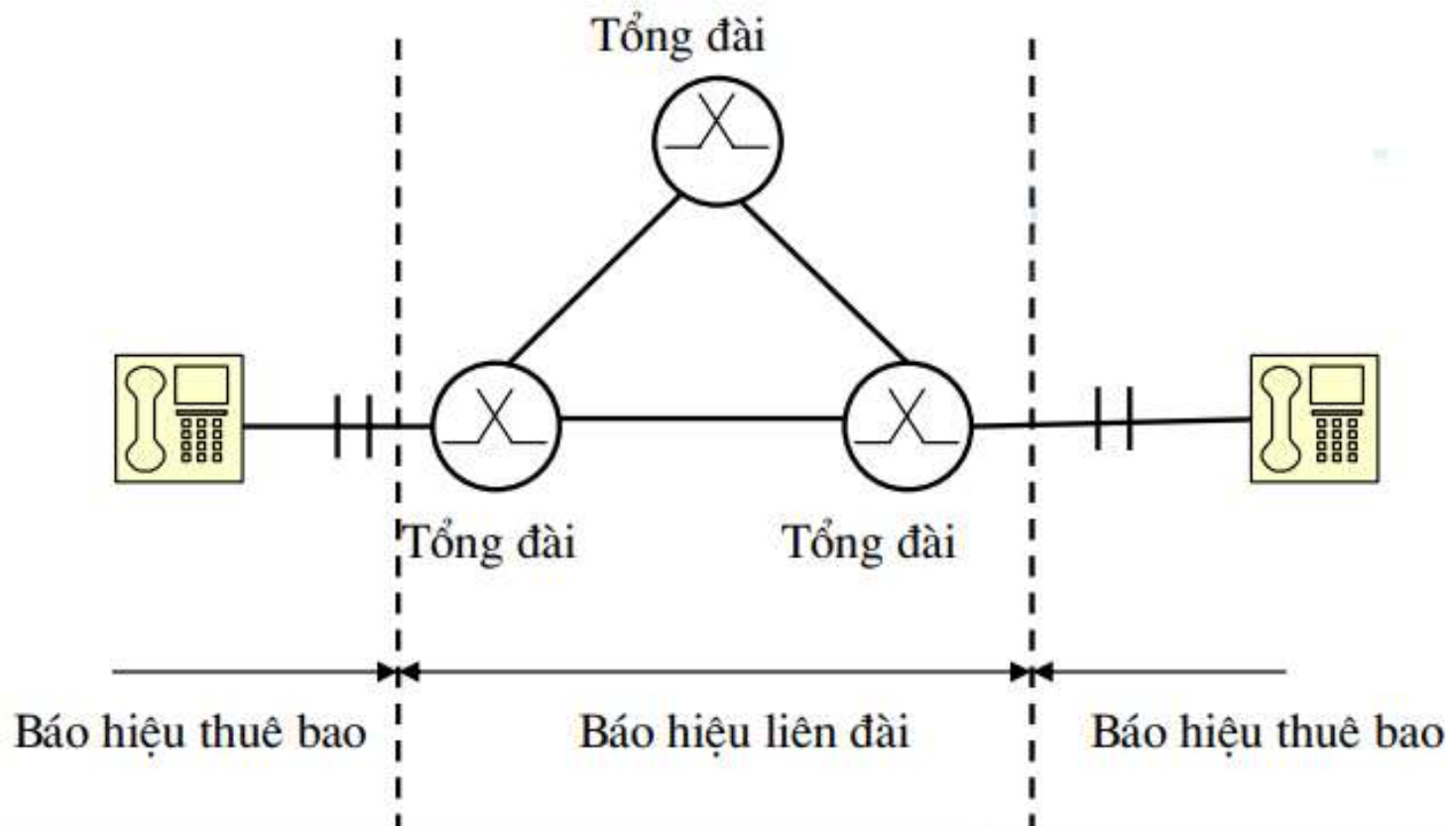
Báo hiệu thuê bao Analog



- Chu trình báo hiệu (thuê bao điện thoại cố định)

Báo hiệu liên đài

Báo hiệu liên đài:



Báo hiệu

Giới thiệu chung về báo hiệu liên đài

- Ở Việt Nam hiện nay sử dụng hai loại hệ thống báo hiệu liên (tổng) đài:
 - Hệ thống báo hiệu kênh kết hợp/kênh riêng R2 (Channel Associated Signalling)
 - Hệ thống báo hiệu kênh chung số 7 CCS7 (Common Channel Signalling number 7) còn gọi là S7 hay SS7 (Signalling System Number 7 - Hệ thống báo hiệu số 7).

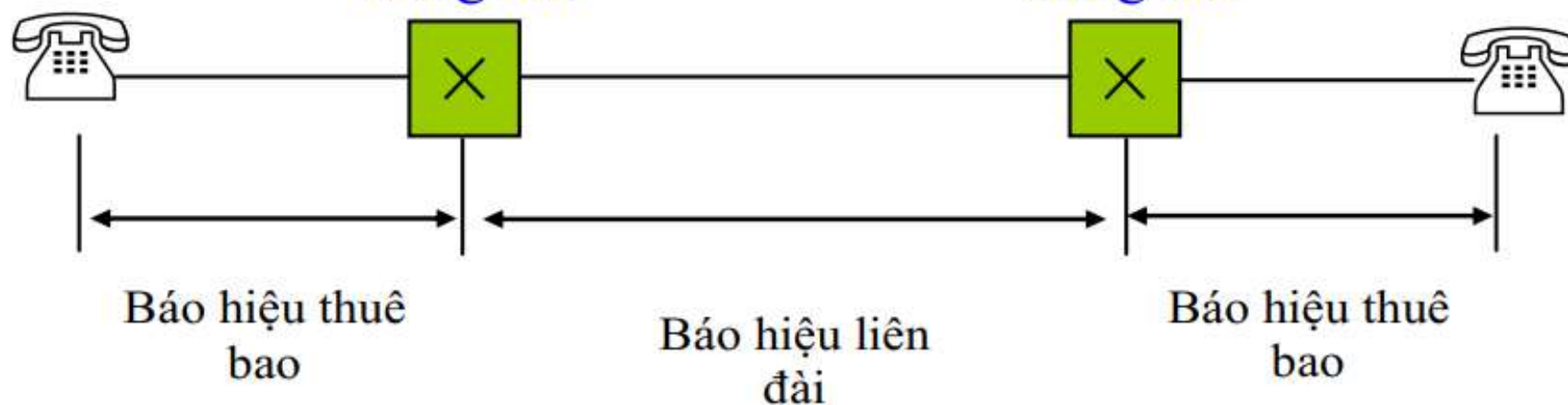
Khái niệm báo hiệu liên đài

Máy điện
thoại

Tổng đài

Tổng đài

Máy điện
thoại



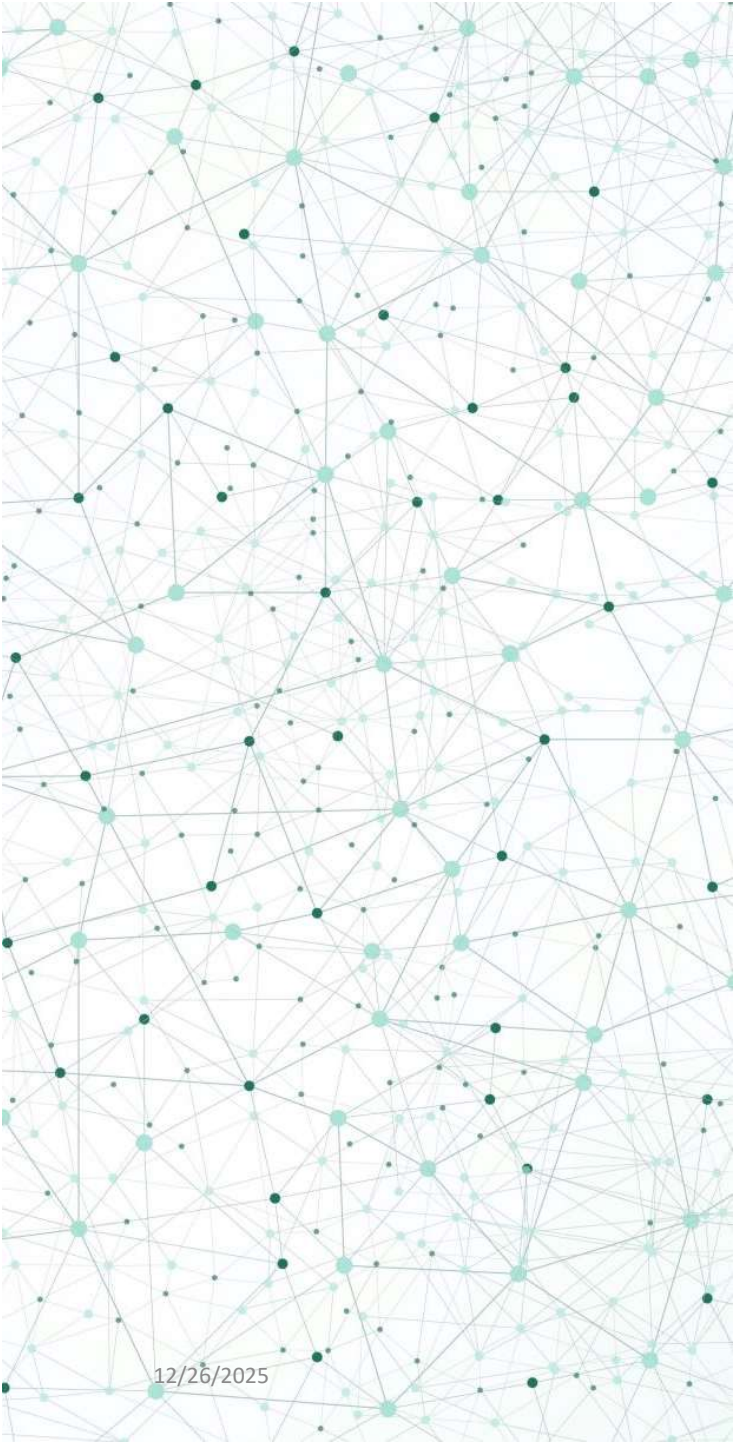
Báo hiệu liên đài được phân chia thành báo hiệu kênh kết hợp (CAS) và báo hiệu kênh chung (CCS). Đối hệ thống báo hiệu kênh kết hợp, các tín hiệu được thu và phát trên cùng một đường với tín hiệu tiếng nói. Trong khi đó, báo hiệu kênh chung tín hiệu báo hiệu được thu và phát qua một đường dành riêng cho báo hiệu khác với kênh tiếng nói.

Hệ thống báo hiệu số 7

- Hệ thống báo hiệu số 7 là hệ thống báo hiệu kênh chung, có nhiệm vụ truyền thông tin báo hiệu giữa các tổng đài với nhau để thiết lập, kết nối, quản lý và giám sát các phiên truyền thông (cuộc gọi thoại, truyền số liệu, hình ảnh...).

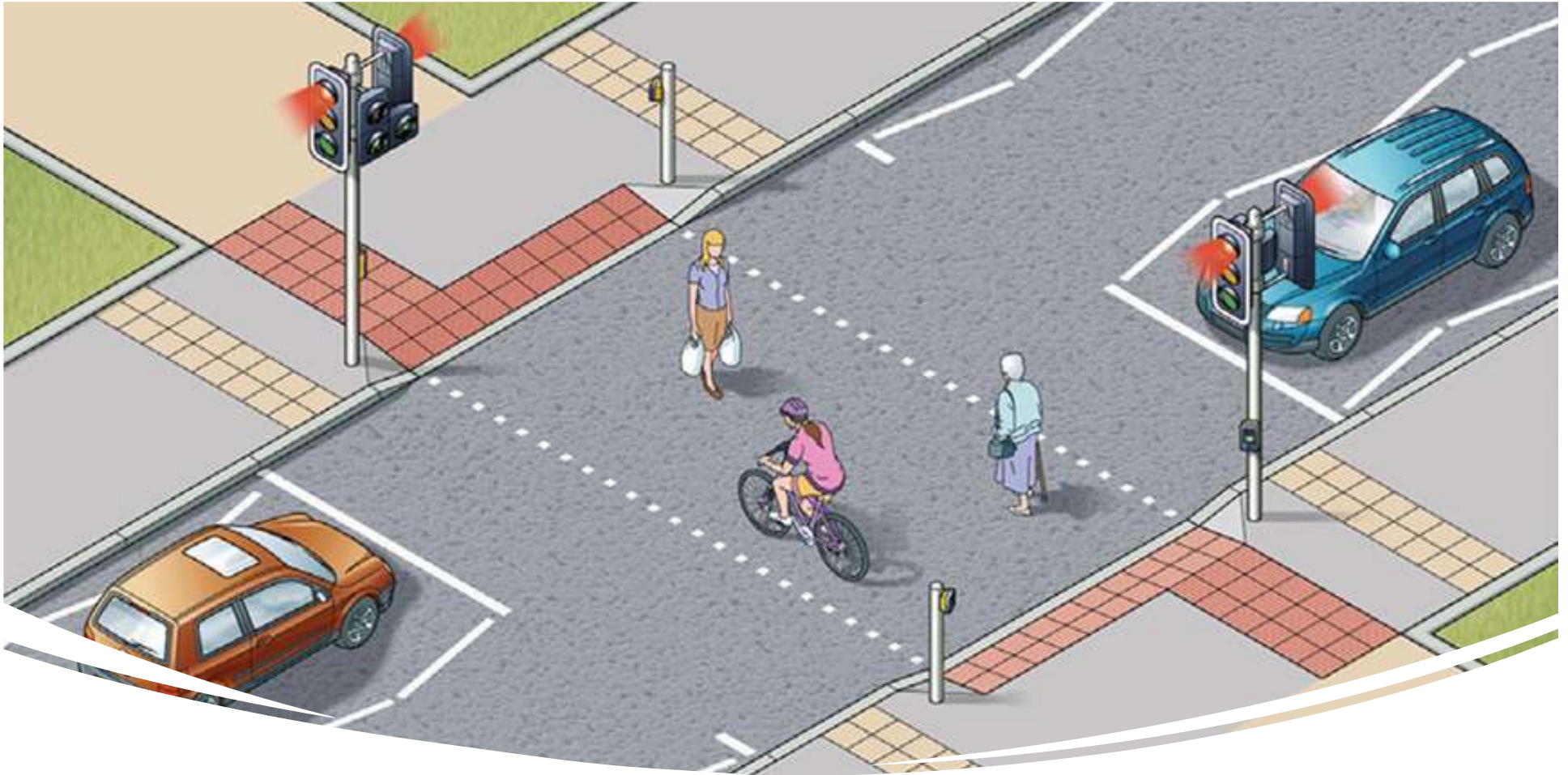
Ưu điểm SS7

- Cấu trúc modun với kiến trúc lớp cho phép nhanh chóng đưa vào các dịch vụ mới.
- Nhanh: Trong hầu hết các trường hợp, thời gian thiết lập cuộc gọi ít hơn 1 giây.
- Dung lượng cao: Một kênh báo hiệu có thể xử lý tín hiệu báo hiệu cho hàng nghìn cuộc gọi cùng lúc.
- Kinh tế: Ít đòi hỏi thiết bị báo hiệu hơn so với các hệ thống khác.
- Tin cậy: Mạng SS7 có độ tin cậy rất cao nhờ sử dụng các tuyến và nút báo hiệu dự phòng. Có khả năng tự động điều chỉnh cấu hình mạng độc lập với mạng thoại.
- Linh hoạt: Hệ thống có thể chứa nhiều tín hiệu, có thể sử dụng cho nhiều mục đích chứ không chỉ dành cho dịch vụ thoại.



Contents

1. Introduction
2. Signaling modes
3. Signaling architecture
4. Protocol Stack
5. Signaling units
6. ISUP and TCAP



1. Introduction

What is signaling?

- Signaling to setup a call (exchange and interact information)
- A setup process is included, call setup, connection, teardown and billing.

What is signaling? (cont.)

- Three categories of signaling information is transmitted:
 - 1) Setup and supervision
 - 2) Service-related information as forwarding, callback, charging
 - 3) Status change information:
 - Transmission network congestion
 - Neighborhood exchange congestion

What is signaling? (cont.)

In order to route telephone traffic through the Public Switched Telephone Network (PSTN), it is necessary to communicate with the switches that make up the PSTN. Signaling is a means for transferring network-related information between switching nodes, and also between the end office switches and their subscribers.

Signaling is used to do the following:

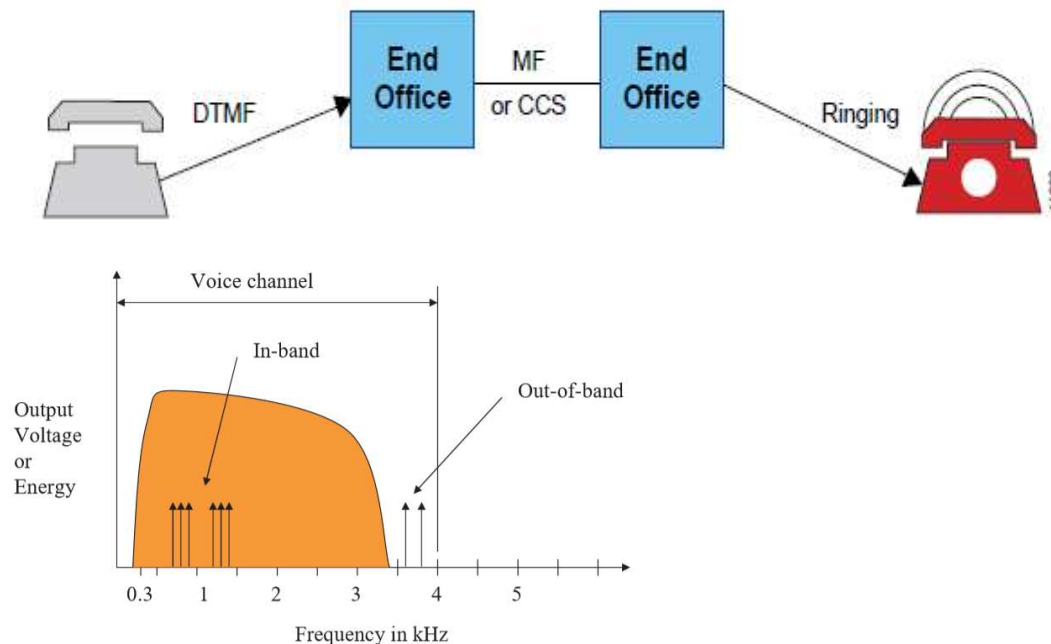
- Request service from the central office switch (via going off-hook).
- Provide central office switch with the information necessary to route a telephone call (via DTMF addressing digits in a specific format).
- Alert destination address of incoming call (ringing).
- Provide status information and call supervision for billing.
- Manage network lines/trunks (set up and teardown calls).

Evolution of signaling techniques

a) Channel Associated Signaling (CAS)

Signaling and data (voice signal and data) are transmitted in the same channel. Ex: subscriber signaling.

Figure 1-1 End-to-End Signaling



Evolution of signaling techniques

a) Channel Associated Signaling (CAS)

When used for in-band signaling:

- Call setup information (off-hook, dialtone, address digits, ringback, busy) is transmitted in the same band of frequencies as used by the voice signal.
- Voice (talk) path is cut over only when the call setup is complete, using the same path that the call setup signals used.
- SF (single-frequency) signaling uses tones to represent on-hook or payphone deposits.
- MF (multi-frequency) signaling is used for switch-to-switch call setup

The principal advantage of CAS is that it is inexpensive to implement and can be used on any transmission medium.

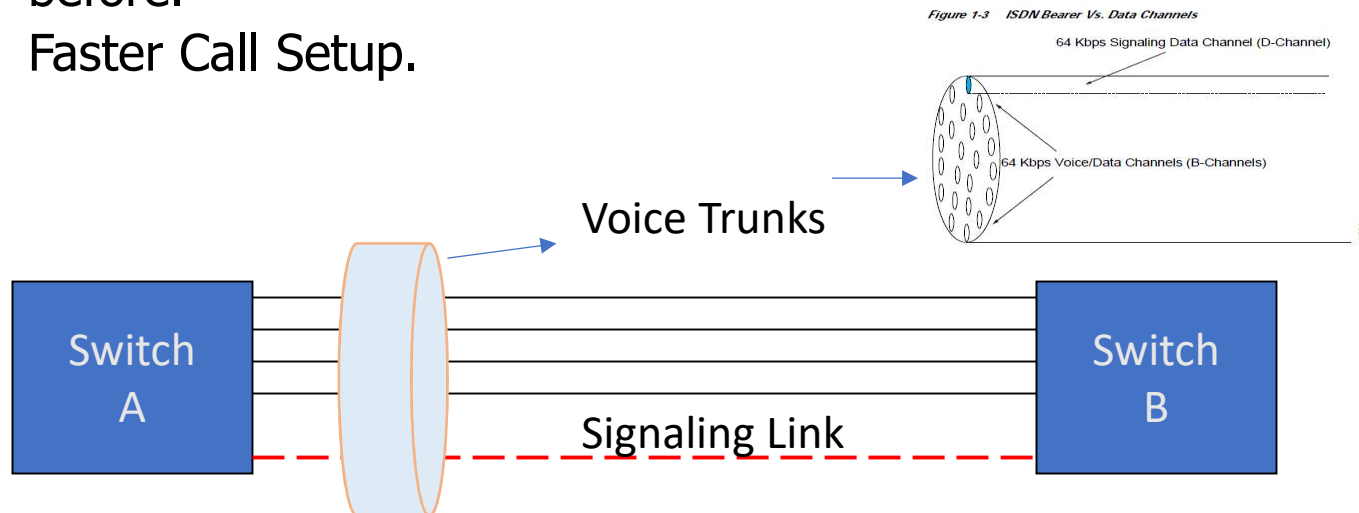
However, CAS has the following disadvantages:

- Fraud— “phone freaks” can build boxes to play call setup and teardown tones.
- Interference is possible between signaling tones used by the network and frequencies of human speech patterns.
- Speed—call setup and teardown is slower, less efficient use of resources.

Evolution of signaling techniques (cont.)

b) Channel Common Signaling (CCS)

- Out of Band signaling
- Many (Maximum of 4096) voice channels Signaling information can be sent on one signaling channel
- Employs separate, dedicated path for signaling.
- Voice trunks are used only when a connection is established, not before.
- Faster Call Setup.



Advantage of CCS over CAS

- Faster call setup
- No interference between signaling tones by network and frequency of human speech pattern.
- Greater Trunk Efficiency: CCS has shorter call set up and tear down times that result in less call holding time, thereby reducing the traffic on the network.
- Variable Information Transfer: CCS allows the transfer of additional information along with the signaling traffic providing facilities such as caller identification and voice or data identification

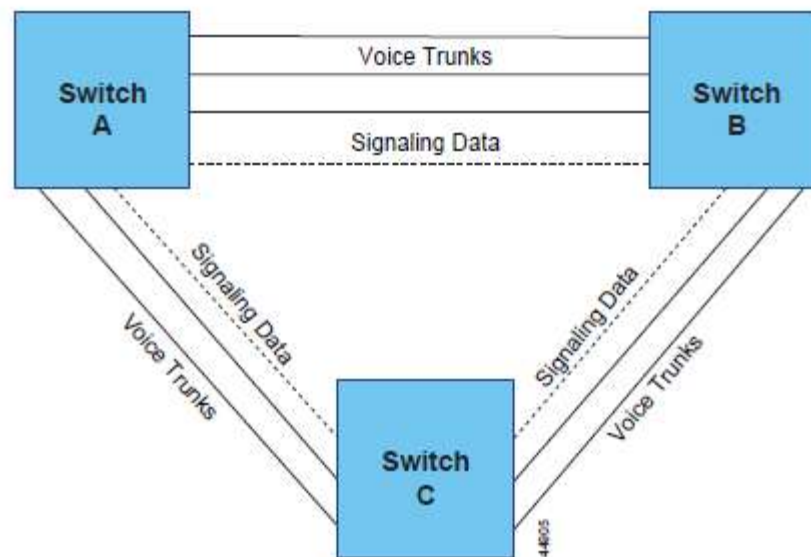
Signaling System 7 (SS7)

- CCITT developed a digital signaling standard called Signaling System 6
- SS6 was based on Packet-Switched, proprietary data network, using 2.4 Kbps data links to send packets of data to distant switches to request service.
- SS7 began deployment in 1983, was initially used for inter office network, but now it is deployed in local central offices.
- Provide a global standard for call setup, routing, control and database access.

2. Signaling Modes of SS7

Associated signaling: Uses one dedicated path between switches as the signaling link. Examples: ISDN-PRI and E1-CAS.

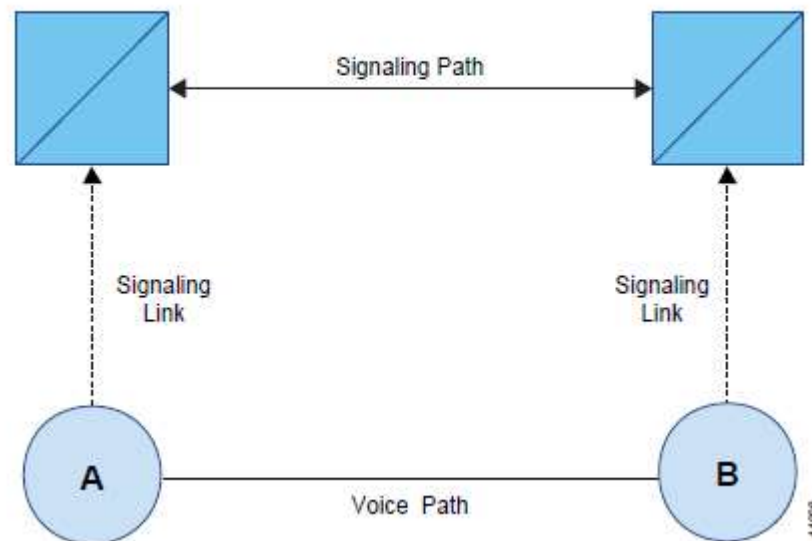
Figure 1-4 Associated Signaling



2. Signaling Modes of SS7 (cont.)

Non-Associated signaling: With this type of signaling, voice/data and signaling are carried on separate, logical paths. Multiple nodes in the signaling path to the final destination can cause delays.

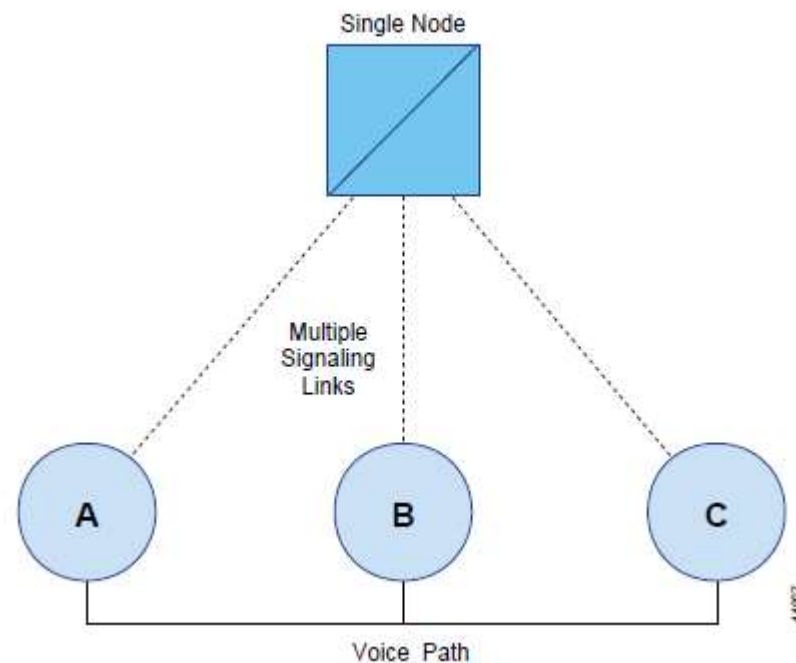
Figure 1-5 Non-Associated Signaling



2. Signaling modes of SS7

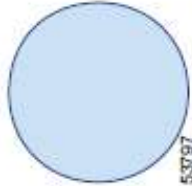
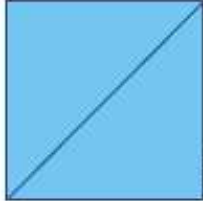
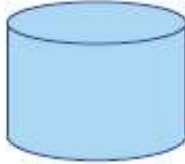
Quasi-Associated Signaling: employs a minimal number of nodes, thus minimizing delays. Quasi-associated signaling is the preferred for SS7.

Figure 1-6 Quasi-Associated Signaling



3. SS7 Signaling Architecture

Table 2-1 SS7 Network Signaling Components

Abbreviation	Name	Symbol
S\$P	Signal Switching Point - or - Service Switching Point	
STP	Signal Transfer Point	
SCP	Signal Control Point - or - Service Control Point	

3. SS7 Signaling Architecture (cont.)

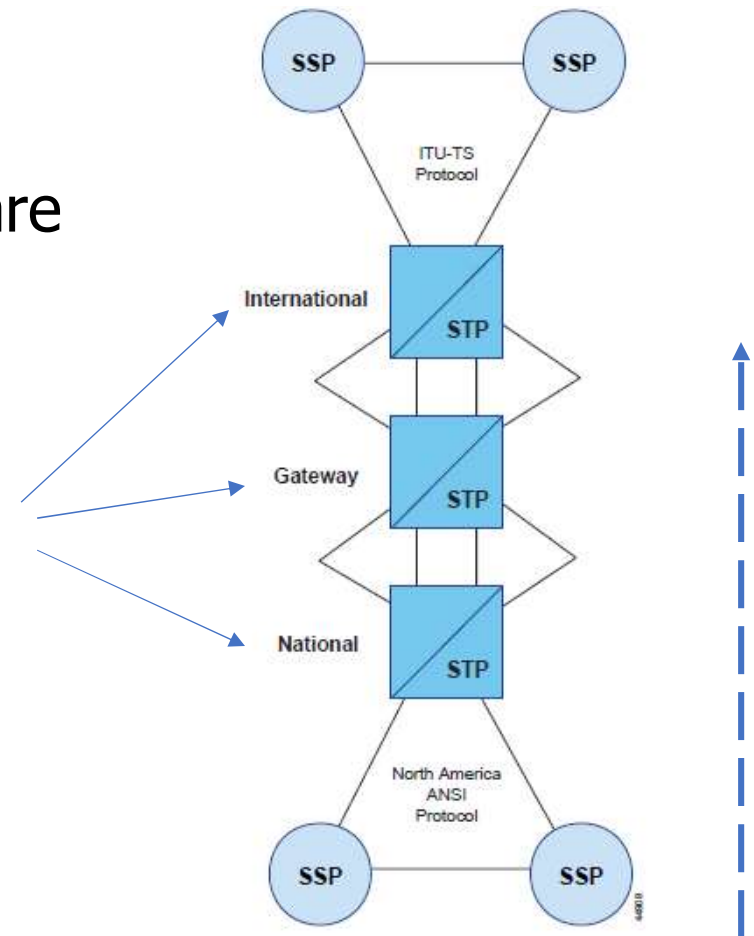
Signal Switching Point (SSP): Computer and software to route and switch calls.

- SSPs create packets and send those signaling messages to other SSPs, as well as queries to remote shared databases to find out how to route calls. They can originate, terminate, or switch calls.
- SSPs communicate with the voice switch via the use of primitives and have the ability to send signaling messages using ISUP (call setup and teardown) and TCAP (database lookup) protocols.

3. SS7 Signaling Architecture (cont.)

Signal Transfer Point (STP) are packet Switches.

There are three levels of STPs



3. SS7 Signaling Architecture (cont.)

Signal Control Point (SCP): An SCP is usually a computer used as a front end to a database system. It is an interface to telco databases, not usually to other, application-specific databases.

- Telco databases are usually linked to SCPs by X.25 links. The SCP can provide protocol conversion from X.25 to SS7 or can provide direct access to the database through the use of primitives which support access from one level of protocol to another.
- The address of an SCP is a point code, and the address of the database it interfaces with is a subsystem number. The database is an application entity which is accessed via the TCAP protocol.

3. SS7 Signaling Architecture (cont.)

Signal Control Point (SSP):

Table 2-2 Telco Databases Accessible via SCP

Abbreviation	Name	Description
BSDB	Business Services Database	Allows companies to create and store proprietary databases, as well as create private networks.
CMSDB	Call Management Services Database	Provides information relating to call processing, network management (prevent congestion), call sampling (create reports for traffic studies), and the routing, billing and third-party billing for 800, 976 and 900 numbers.
HLR	Home Location Register	Used in cellular networks to store subscriber information.
LIDB	Line Information Database	Provides billing instructions.

3. SS7 Signaling Architecture (cont.)

Signal Control Point (SSP):

Table 2-2 Telco Databases Accessible via SCP (continued)

Abbreviation	Name	Description
LNP	Local Number Portability	Allows people to change telco service providers but keep their same telephone number.
OSS	Operations Support Systems	Associated with remote maintenance centers for monitoring and managing SS7 and voice networks.
VLR	Visitor Location Register	Used when a cell phone is not recognized by the mobile switching center (MSC).

3. SS7 Signaling Architecture (cont.)

SS7 Links:

An SS7 link is the physical transmission line (serial 56/64 Kbps or DS0 channel) that connects the individual nodes in an SS7 network.

SS7 networks are built to be highly reliable and redundant. Link diversity is built into the network design, providing multiple signaling paths, so that there is no single point of failure. This practice ensures that redundant links have the capacity to handle all rerouted network traffic.

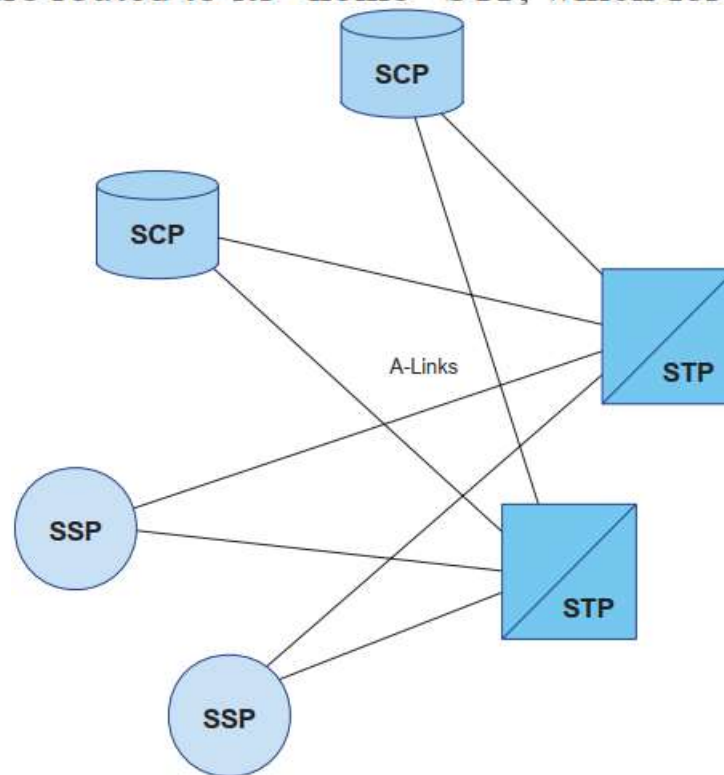
A-Links:

3. SS7 Signaling Architecture (cont.)

A-Links:

Access links (A-links) interconnect an STP and either an SSP or an SCP (signaling end points). Their sole purpose is to deliver signaling to and from signaling end points. End points always have at least two A-links (also called signaling beginning points).

Any signaling that an SSP or SCP needs to send to any other node in the SS7 network is sent on one of its A-links to its “home” STP, which processes and routes the message along its way. Messages addressed to an SSP or SCP are routed to its “home” STP, which forwards them to the addressed node over its A-links.

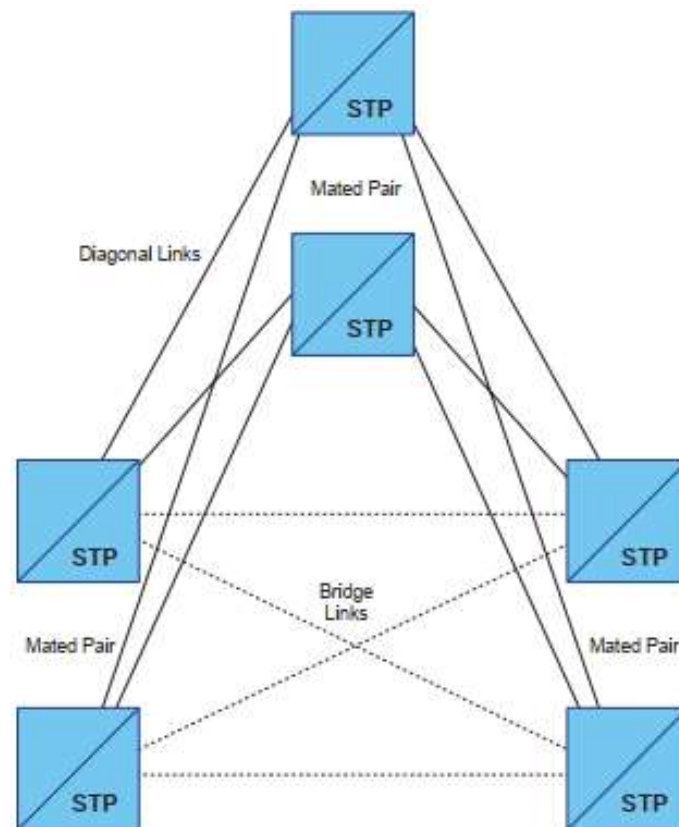


3. SS7 Signaling Architecture (cont.)

B-and D-Links:

Bridge links (B-links) are the quad of links interconnecting peer pairs of STPs. Diagonal links (D-links) are the quad of links interconnecting mated pairs of STPs at different hierarchical levels. (See Figure 2-3.)

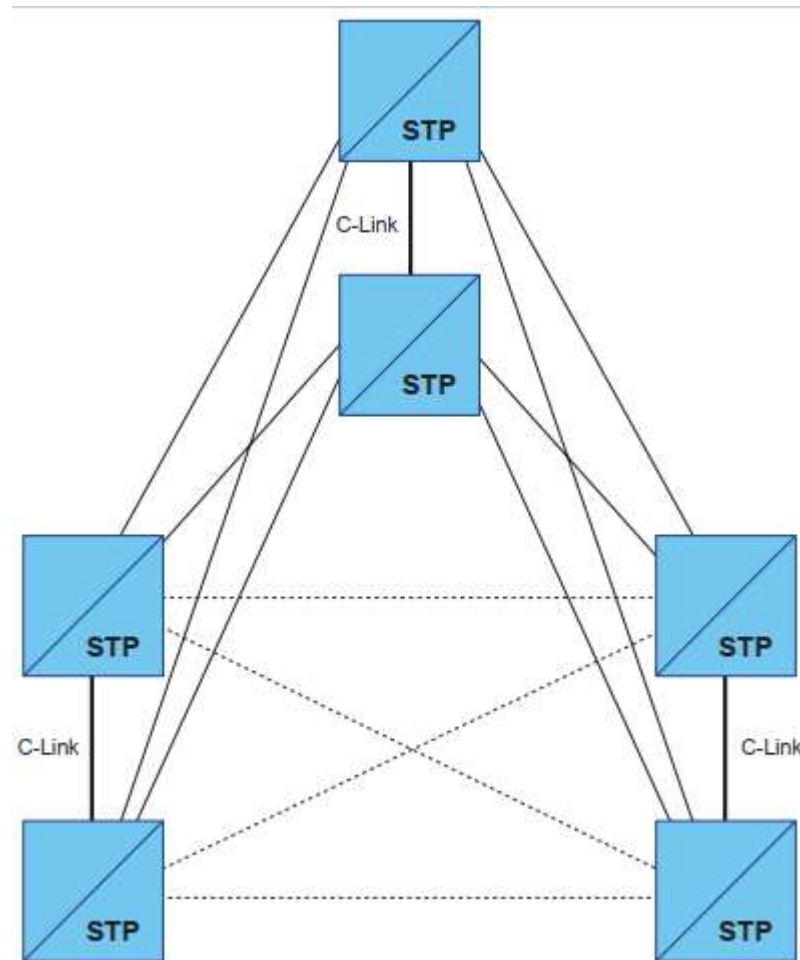
Since the SS7 network has no clear hierarchy, these links are referred to as B-links, D-links, or B/D-links.



3. SS7 Signaling Architecture (cont.)

C-Links:

Cross links (C-links) interconnect mated STPs and are used to enhance the reliability of the signaling network not regularly used by SS7 traffic. (See Figure 2-4.) They are used only when there has been a link failure which causes an STP to have no other route.



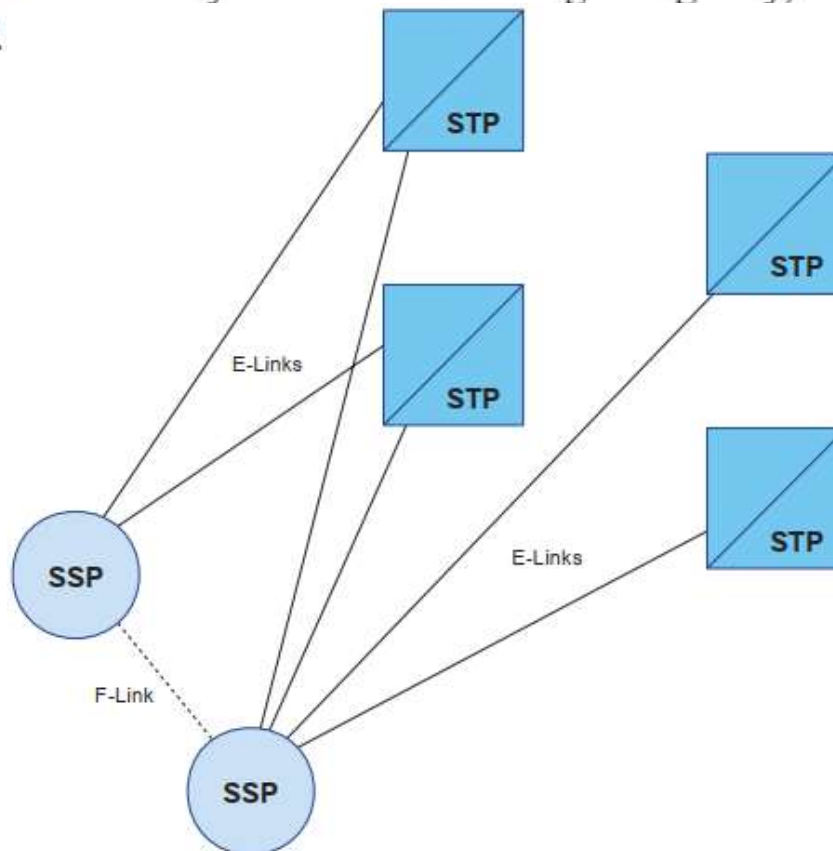
3. SS7 Signaling Architecture (cont.)

E-and F-Links:

Extended links (E-links) connect an SSP to an alternate STP to provide backup connectivity to the network if the SSP's "home" STP cannot be reached on its A-link.

E-links are not usually provisioned, unless cost/reliability trade-offs justify the expense.

Fully associated links (F-links) directly connect two signaling end points (SSPs and/or SCPs). They are not usually used in networks with STPs because they allow associated signaling only, thus bypassing the security features provided with an STP.

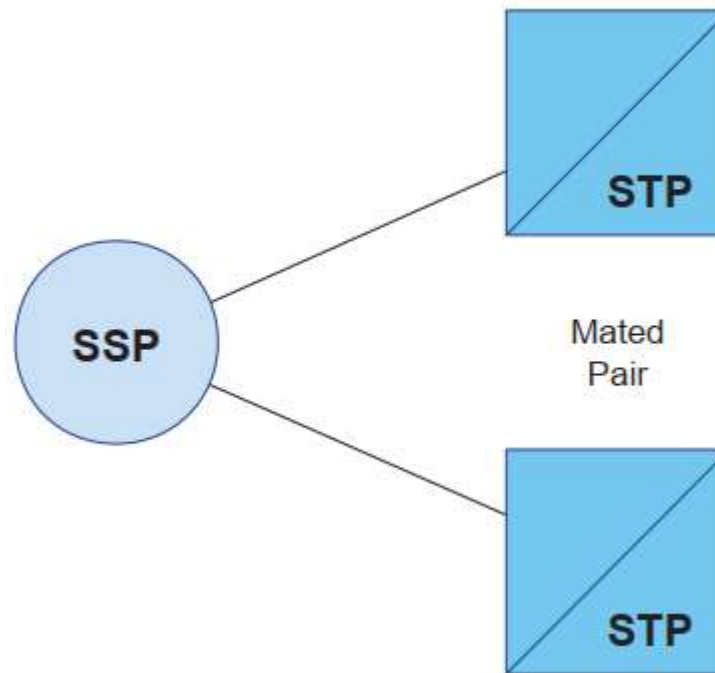


3. SS7 Signaling Architecture (cont.)

Linksets:

Links are put into groups called linksets. Up to 16 links can be assigned to one linkset. All links in a linkset must have the same adjacent node. (See Figure 2-6.)

Switches will alternate traffic across all links in a linkset to ensure equal usage of all facilities in the network.



3. SS7 Signaling Architecture (cont.)

Linkset characteristics:

If possible, links should be terrestrial. Satellite links can be used but are not preferred because of the inherent delay.

Alternate linksets are set up to provide backup paths when congestion occurs in the network. When a link fails, all other links within the linkset must take over. (See Figure 2-7.)

A maximum of 10 minutes downtime per year is allowed for any one linkset, to protect network integrity.

If an SS7 entity such as an STP fails, its mate assumes the full traffic load. For this reason, SS7 entities are designed to send less than 40 percent of the traffic on any given link. If an entity fails at 40-percent capacity, there is still enough room on its mate for it to carry the entire traffic load of the mated pair.

3. SS7 Signaling Architecture (cont.)

Linkset Interfaces:

The signaling link interface type will depend on the type of equipment used with the links. The V.35 interface is used to connect from the data service unit (DSU) to the signaling point. V.35 can also be used from a digital system cross-connect frame (DSX).

V.35 needs a clock source. Data links are 56 or 64 Kbps.

The most commonly used interface is a DS0A, one 56/64 Kbps channel of a DS1. A channel service unit (CSU) or DSU terminates the DS1 and separates DS0s from the T1 or E1 span circuit.

3. SS7 Signaling Architecture (cont.)

Routes:

The signaling point must define linksets and routes in SS7 messaging. The following entities are used in SS7 messaging:

- **Route**—A collection of linksets to reach a particular destination. A linkset can belong to more than one route.
- **Routeset**—A collection of routes that are assigned to destinations and also provide alternate routes.
- **Destination**—An address entered into the routing table of a remote signaling point. A destination need not be adjacent to the signaling point, but must be a point code that can be reached by the signaling point.

3. SS7 Signaling Architecture (cont.)

Point Codes:

In SS7, addresses are assigned using a three-level hierarchy.

- **Member**—A signaling point within a cluster.
- **Cluster**—A collection of signaling points (members).
- **Network**—Each cluster is defined as being part of a network.

Any node in the SS7 network can be addressed by the three-level number defined by its network, cluster, and member numbers. Each of these numbers is an 8-bit number assigned a value from 0 to 255. This three-level address is called the *point code* of the signaling point.

3. SS7 Signaling Architecture (cont.)

Network Numbers:

Network numbers are assigned on a nationwide basis. In North America, RBOCs, IXC's and telcos already have network numbers assigned to them.

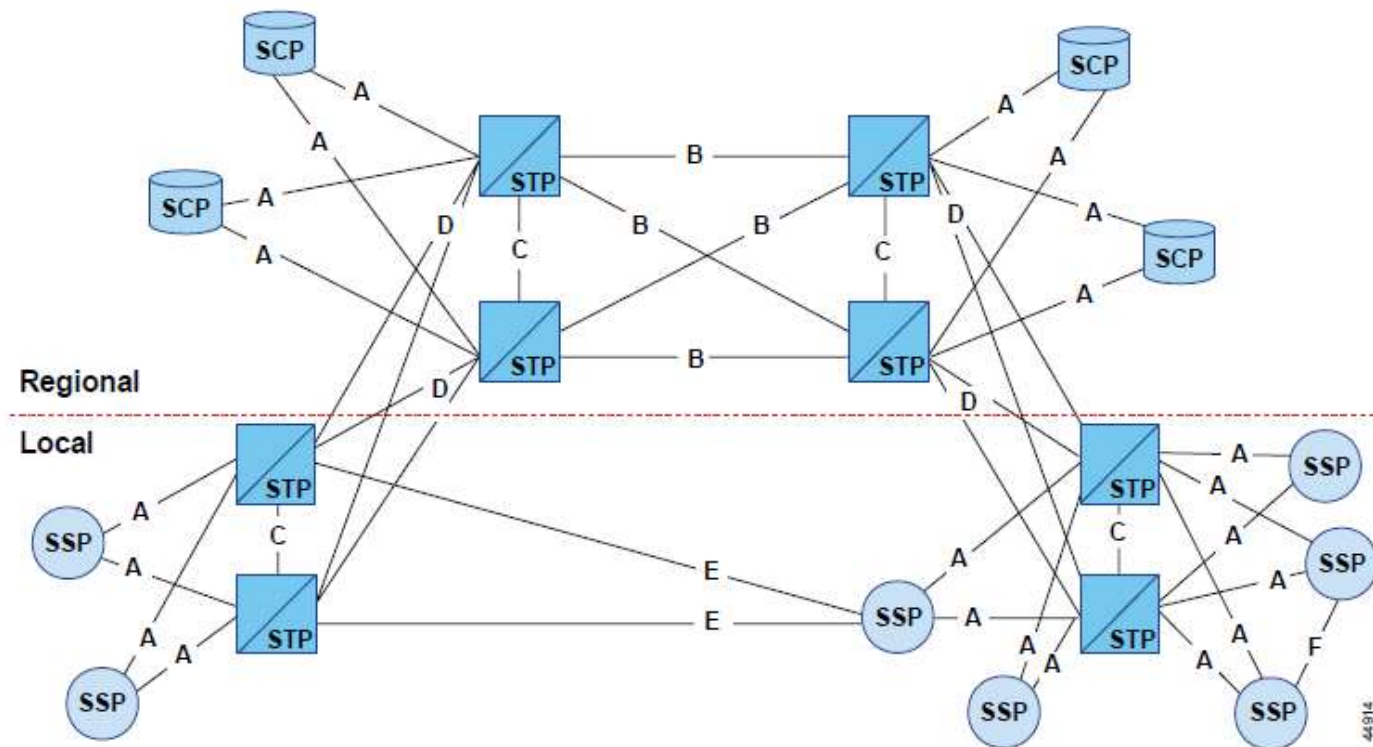
Network numbers are relatively scarce. Companies are expected to meet size requirements in order to be assigned a network number.

Network number 0 is not available and 255 is reserved.

Smaller networks can be assigned one or more clusters within network numbers 1, 2, 3 and 4. The smallest networks are assigned point codes within network number 5. The cluster to which they are assigned determines the state or province they are in.

3. SS7 Signaling Architecture (cont.)

Figure 2-7 SS7 Network



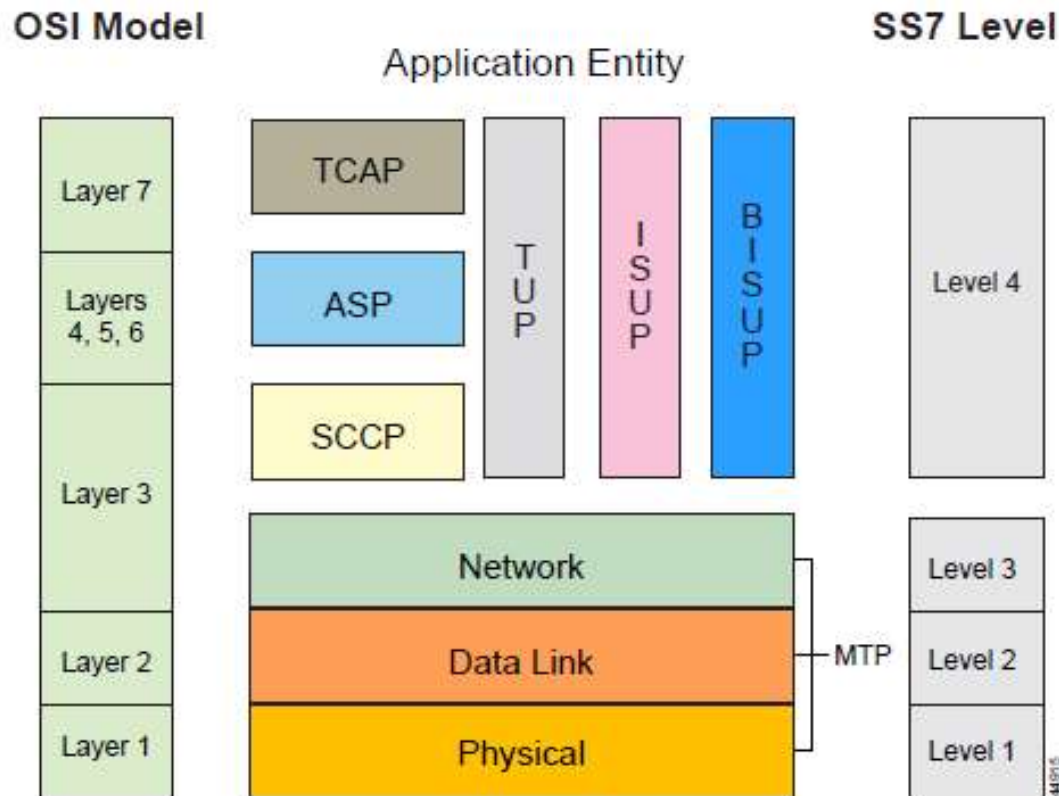
3. SS7 Signaling Architecture (cont.)

SS7 Links:

- How many type of links in SS7 network?
- Why do they define many type of links in SS7?
- Why are SS7 networks built to be reliable and redundant?
- How to built a reliable SS7 network?

4. SS7 Protocol Stack

This chapter describes the components of the SS7 protocol stack. A stack is a set of data storage locations that are accessed in a fixed sequence. The SS7 stack is compared against the Open Systems Interconnection (OSI) model for communication between different systems made by different vendors.



TCAP: Transaction Capabilities Application Part
ASP: Application Service Part
SCCP: Signaling Connection Control Part
TUP: Telephone User Part
ISUP: ISDN User Part
BISUP: Broadband ISDN User Part
MTP: Message Transfer Part

4. SS7 Protocol Stack

Level 1: Physical Connection

This is the physical level of connectivity, virtually the same as Layer 1 of the OSI model. SS7 specifies what interfaces will be used, both Bellcore (Telecordia) and ANSI call for either the DS0A or the V.35 interface.

Because central offices are already using DS1 and DS3 facilities to link one another, the DS0A interface is readily available in all central offices, and is preferred in the SS7 network. As the demands on the SS7 network increase (local number portability), and as the industry migrates toward ATM networks, the DS1 interface will become the link interface.

4. SS7 Protocol Stack

SS7 Level 2: Data Link

The data link level provides the network with sequenced delivery of all SS7 message packets. Like the OSI data link layer, it is only concerned with the transmission of data from one node to the next, not to its final destination in the network.

Sequential numbering is used to determine if any messages have been lost during transmission. Each link uses its own message numbering series independent of other links.

SS7 uses CRC-16 error checking of data and requests retransmission of lost or corrupted messages. Length indicators allow Level 2 to determine what type of signal unit it is receiving, and how to process it.

4. SS7 Protocol Stack

SS7 Level 3: Network Level

The network level depends on the services of Level 2 to provide routing, message discrimination and message distribution functions.

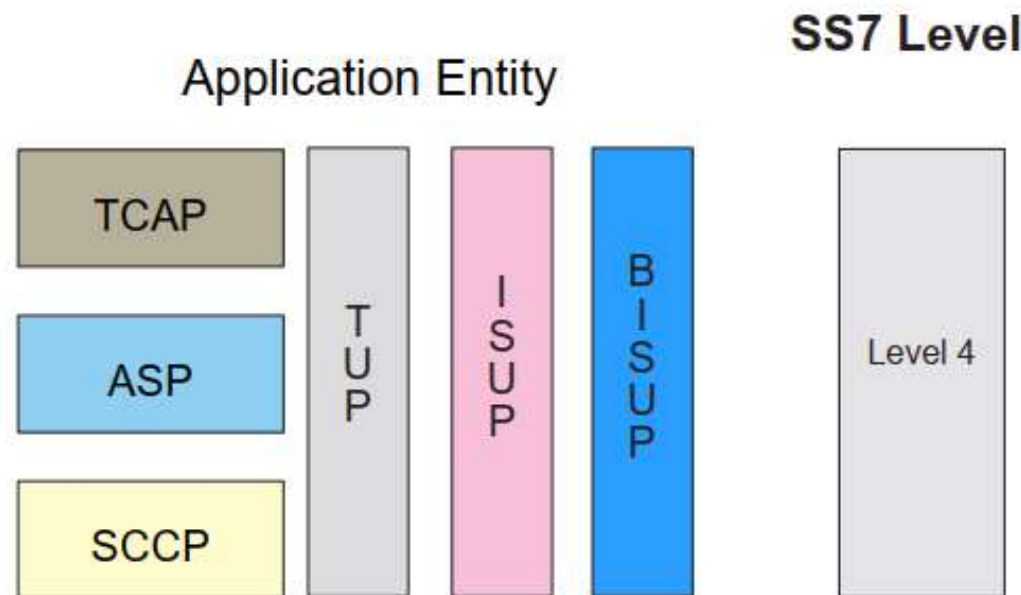
- Message Discrimination determines to whom the message is addressed.
- Message Distribution is passed here if it is a local message.
- Message Routing is passed here if it is not a local message.

4. SS7 Protocol Stack

SS7 Level 4: Protocols, User and Application Parts

Level 4 consists of several protocols, user parts and application parts. (See Figure 3-3.)

Figure 3-3 SS7 Level 4 Protocols, User and Application Parts



5. SS7 Signal Unit

Signal Unit Structure

All types of signal units (MSU, LSSU, FISU) have a set of common fields which are used by MTP Level 2. Field types include the following:

- **Flag**—Delimiter in a signal unit which marks the end of one signal unit and the beginning of another. All signal units begin with a distinct 8-bit pattern (0111 1110).



Note

Although the protocol allows an opening and closing flag, only one flag is used in North America.



- **Checksum**—An 8-bit sum calculated from the transmitted message by the transmitting signaling point and inserted in the message. It is recalculated by the receiving signaling point, and if corrupted, a retransmission is requested.
- **Length Indicator**—The number of octets between itself and the checksum. Checks the integrity of the signal unit and discriminates between different types of signal units. The default values are: FISU=0, LSSU=1 or 2, MSU>2
- **BSN/BIB FSN/FIB**—Octets that hold the backward sequence number (BSN) and backward indicator bit (BIB); the forward sequence number (FSN) and the forward indicator bit (FIB).

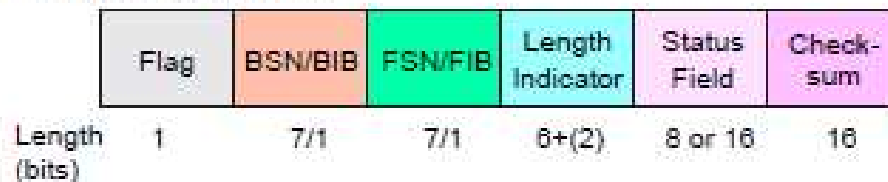
5. SS7 Signal Unit

First bit transmittal ← Last bit transmittal

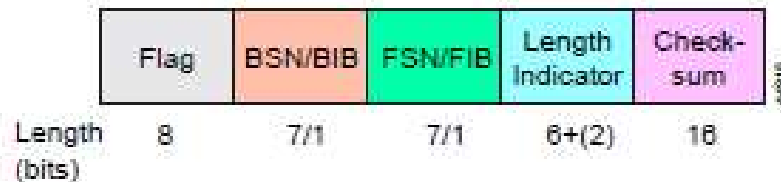
Message Signal Unit



Link Status Signal Unit



Fill-In Signal Unit

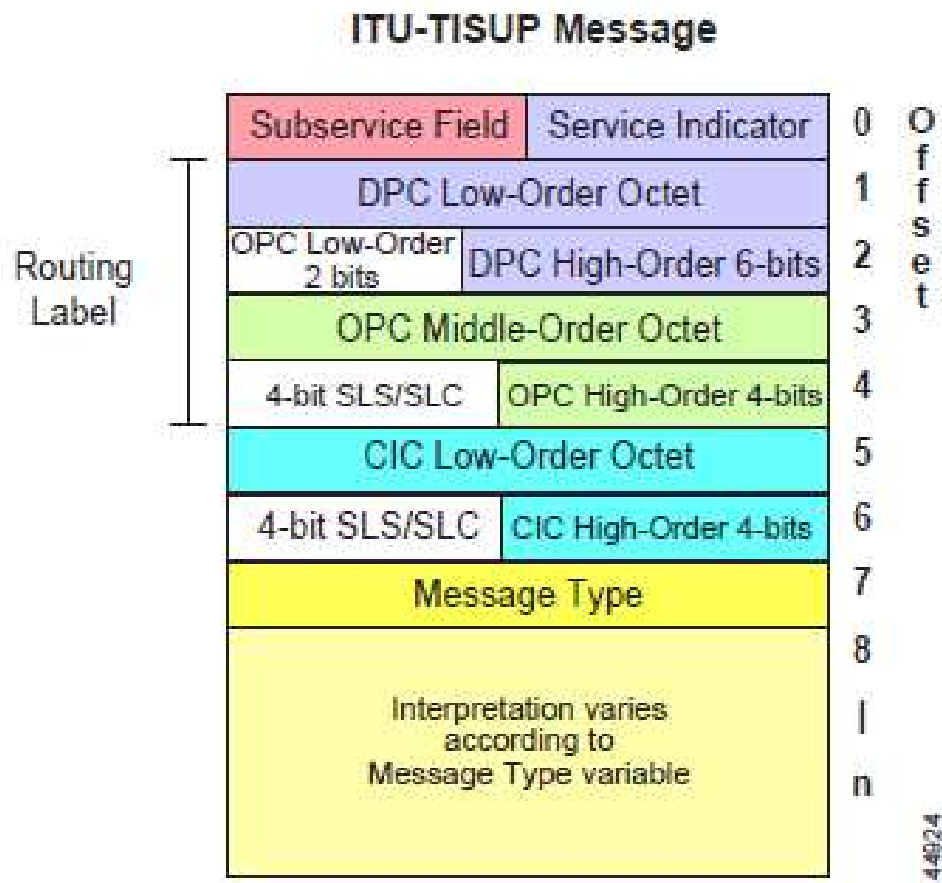


- ❖ What is signal unit ?
- ❖ How are the meaning of SU types?
- ❖ Define the position layer of fields
- ❖ How are you clear about SIO/SIF?

Table 4-1 SIO Service Indicator Bits

Value	Function
0	Signaling Network Management
1	Signaling Network Testing and Maintenance
2	Signaling Connection Control Part (SCCP)
3	ISDN User Part (ISUP)

6. ISUP and TCAP



- ❖ What are the functions of ISUP/TCAP?
- ❖ How are the meaning of SU types?
- ❖ Define the position layer of fields
- ❖ How are you clear about SIO/SIF?

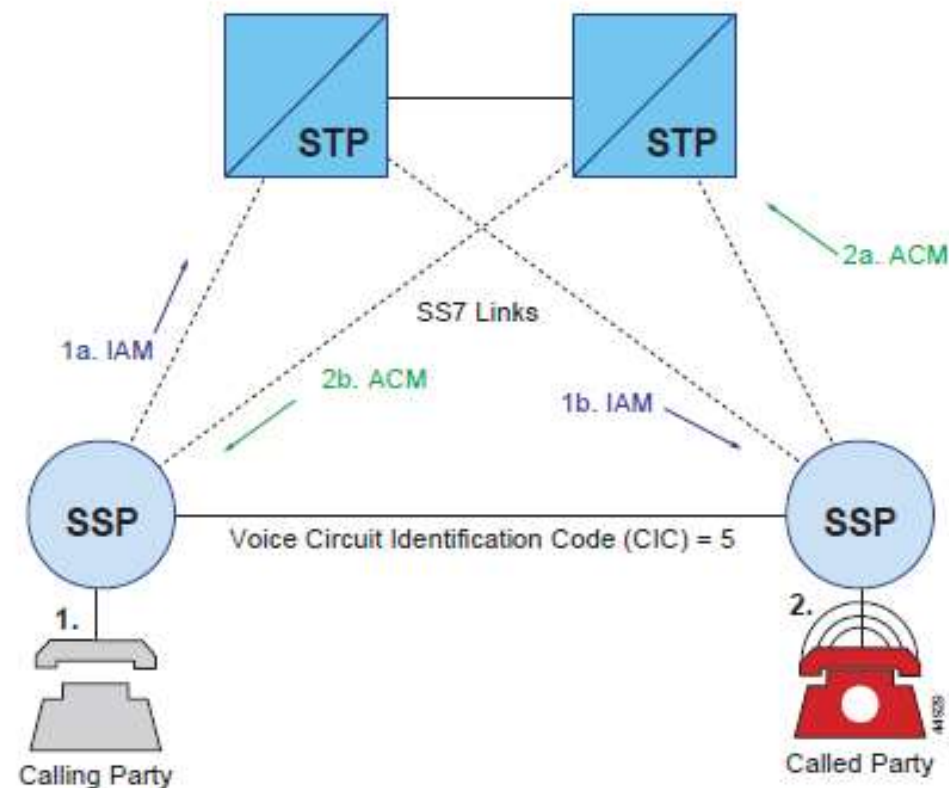
6. ISUP and TCAP (cont.)

Basic ISUP Signaling

ISDN User Part (ISUP) defines the protocol and procedures used to set up, manage and release trunk circuits that carry voice and data calls over the PSTN. ISUP is used for both ISDN and non-ISDN calls. Calls that terminate within the same switch do not use ISUP signaling. (See Figure 5-1.)

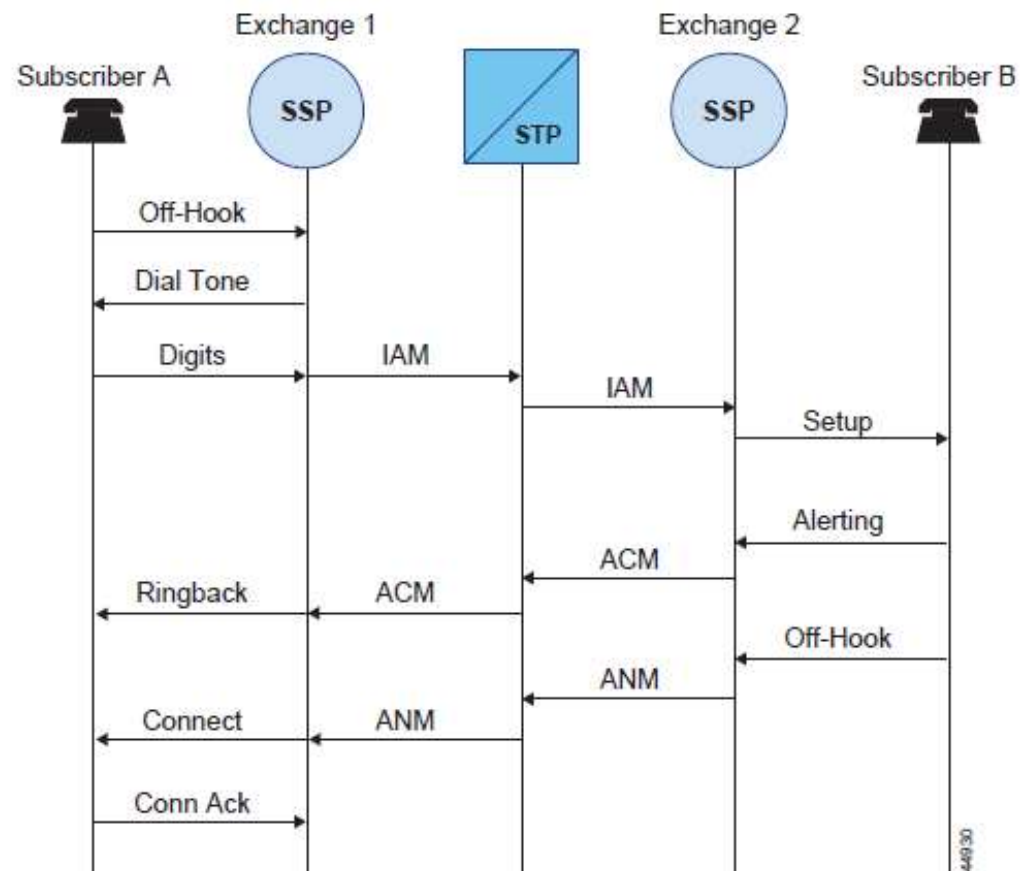
In some parts of the world, such as China, the Telephone User Part (TUP) protocol supports basic call processing. TUP handles analog circuits only; digital circuits and data transmission capabilities are supported by the Data User Part protocol.

Figure 5-7 ISUP Call Initiation (1)



6. ISUP and TCAP (cont.)

Figure 5-8 ISUP Call Initiation (2)



Exercises

1. Name the types of signaling used in PTSN.
2. Which signaling types categorizes SS7?
3. Compare the advantages and disadvantages between CCS and CAS.
4. Name these modes of CCS and what mode is referred to SS7? Why?
5. Compare the advantage between SS7 and others signaling forms.
6. Draft a basic architecture of SS7 network and describe briefly the function of each modules.
7. Present the SS7 protocol stack in comparison with OSI model and describe the function of each modules.
8. Name the three types of signal units and describe briefly their functions.

1. A 400 km repeater system consists of m identical cable sections with $\alpha = 0.4$ dB/km and m identical amplifiers with 30 dB maximum gain. Find the required number of sections and the gain per amplifier so that $P_{\text{out}} = 50$ mW when $P_{\text{in}} = 2$ W.

Design a SDS switch of a PABX (Private Automatic Branch Exchange). We assumed that the PABX serves for 125 subscribers with blocking probability at 0.001 and utilization probability is 0.3 at the busy hour.